PHYS 372 Formula Sheet

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Schrodinger Equation

Time-Dependent Schrodinger Equation

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x,t)\right]\Psi(x,t)$$

Time-Independent Schrodinger Equation

$$\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x) = E\psi(x) \qquad \Psi(x,t) = e^{\frac{-iE}{\hbar}t}\psi(x)$$

Basic Identities

$$[\hat{x},\hat{p_x}] = i\hbar \qquad m\frac{d}{dt}\langle x\rangle = \langle p\rangle \qquad \frac{d}{dt}\langle p\rangle = -\langle V'(x)\rangle$$

$$[\hat{x},\hat{H}] = \frac{i\hbar}{m}\hat{p} \qquad \frac{d}{dt}\langle\hat{A}\rangle = \frac{1}{i\hbar}\langle[\hat{A},\hat{H}]\rangle + \left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle$$

Free Particle

Eigenfunctions

$$\Psi_k(x,t) = Ae^{i\left(kx-\frac{\hbar k^2}{2m}t\right)} \qquad k = \pm\frac{\sqrt{2mE}}{\hbar} \qquad k\in\mathbb{R}$$

Free-particle Wavepackets

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\phi(k)e^{i\left(kx-\frac{\hbar k^2}{2m}t\right)}dk \qquad \omega = \frac{\hbar k^2}{2m}$$

Fourier Transform: $\phi(k) = \dfrac{1}{\sqrt{2\pi}}\displaystyle\int_{-\infty}^{\infty}\Psi(x,0)e^{-ikx}dk$

$$\Delta k << k_0 \implies \Psi(x,t)\approx\psi(\frac{\hbar k_0}{m}t - x) \qquad v_g = \frac{\hbar k_0}{m} = \frac{d\omega}{dk}$$

Gaussian Wavepackets

$$\Psi(0,t) = \left(\frac{1}{2\pi\Delta}\right)^{\frac{1}{4}}e^{\frac{(x-x_0)^2}{(2\Delta)^2}+ik_0x} \qquad |\Psi(0,t)|^2\sim\text{Norm}(x_0,\Delta^2)$$

$$\phi(k) = \frac{\sqrt{2}\Delta^{1/2}}{(2\pi)^{1/4}}e^{-(k-k_0)^2\Delta^2-ix_0(k-k_0)} \quad |\phi(k)|^2\sim\text{Norm}(k_0,\frac{1}{4\Delta^2})$$

Char. Func. of Gaussian $\langle e^{itX}\rangle = e^{it\mu-\frac{1}{2}\sigma^2t^2}$

Infinite Square Well

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right) \qquad k_n = \frac{\sqrt{2mE_n}}{\hbar} \qquad E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

Harmonic Oscillator

$$\hat{H}\psi_n = \hbar\omega\left(n+\frac{1}{2}\right) \qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right) \qquad V(x) = \frac{1}{2}m\omega^2x^2$$

Ladder Operators

$$\hat{a}_+ = \frac{1}{(2\hbar m\omega)^{1/2}}(m\omega\hat{x} - i\hat{p}) \qquad \hat{a}_+ = \sqrt{n+1}\psi_{n+1}$$

$$\hat{a}_- = \frac{1}{(2\hbar m\omega)^{1/2}}(m\omega\hat{x} + i\hat{p}) \qquad \hat{a}_-\psi_n = \sqrt{n}\psi_{n-1}$$

$$\hat{H} = \hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right) \qquad [\hat{a}_-,\hat{a}_+] = 1 \qquad \hat{a}_-^\dagger = \hat{a}_+$$

$$\psi_n = \frac{1}{\sqrt{n!}}(a_+)^n\psi_0 \qquad \psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-\frac{m\omega}{2\hbar}x^2}$$

Hermite Polynomials

$$\begin{aligned}
&H_0 = 1,\\
&H_1 = 2\xi,\\
&H_2 = 4\xi^2 - 2,\\
&H_3 = 8\xi^3 - 12\xi,\\
&H_4 = 16\xi^4 - 48\xi^2 + 12,\\
&H_5 = 32\xi^5 - 160\xi^3 + 120\xi.
\end{aligned}$$

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2} \qquad \xi = \sqrt{\frac{m\omega}{\hbar}}x$$

$$H_{n+1} = 2\xi H_n - 2nH_{n-1} \qquad H_n(\xi) = (-1)^n e^{\xi^2}\left(\frac{d}{d\xi}\right)^n e^{-\xi^2}$$

Delta Well Potential

$V(x) = -\alpha\delta(x)$ where $-\infty < x < \infty$

Bound State (Negative E)

$$\psi_1 = \sqrt{\frac{m\alpha}{\hbar^2}}e^{-\frac{m\alpha|x|}{\hbar^2}} \qquad \kappa = \frac{m\alpha}{\hbar^2} \qquad E = -\frac{\hbar^2}{2m}\kappa^2$$

Scattering States (Positive E)

$$E_k = \frac{\hbar^2k^2}{2m} \qquad \psi(x) = \begin{cases} A(e^{ikx} + \dfrac{i\beta}{1-i\beta}e^{-ikx}), & x<0 \\ \dfrac{A}{1-i\beta}e^{ikx}, & x>0 \end{cases}$$

$$\beta = \frac{m\alpha}{\hbar^2k} \qquad R = \frac{|B|^2}{|A|^2} = \frac{\beta^2}{1+\beta^2} \qquad T = \frac{|C|^2}{|A|^2} = \frac{1}{1+\beta^2}$$

Single-Particle Normalization / Scattering Matrix

$$A = \frac{\sqrt{1+\beta^2}}{\sqrt{2\pi}} \qquad S = \frac{1}{1-i\beta}\begin{pmatrix}1 & i\beta\\ i\beta & 1\end{pmatrix} \qquad S^{-1} = S^\dagger$$

Finite Square Well

$V(x) = -V_0$ if $-a < x < a$, otherwise 0

$$\psi(x) = \begin{cases} Ae^{-kx}, & x>a \\ C\sin(lx) + D\cos(lx), & -a<x<a \\ Be^{kx}, & x<-a \end{cases}$$

$$k = \frac{\sqrt{-2mE}}{\hbar} \qquad l = \frac{\sqrt{2m(V_0+E)}}{\hbar}$$

Even and Odd Solutions

$$\psi_e(x) = \begin{cases} Ae^{-kx}, & x > a \\ D\cos(lx), & 0 < x < a \\ \psi_e(-x), & x < -a \end{cases} \qquad \tan(la) = \frac{k}{l} \qquad n = 1, 3, ...$$

$$\psi_o(x) = \begin{cases} Ae^{-kx}, & x > a \\ C\sin(lx), & 0 < x < a \\ -\psi_o(-x), & x < -a \end{cases} \qquad \tan(la) = \frac{-l}{k} \qquad n = 2, 4, ...$$

Deep Well $\implies E_n = -V_0 + \frac{\hbar^2\pi^2n^2}{8a^2m}$

Shallow Well $\implies E \approx \frac{-V_0}{1+\frac{\hbar^2}{2ma^2V_0}} \qquad \frac{2ma^2}{\hbar^2}(E+V_0) << 1$

Transmission and Reflection Coefficients

$$T = 1 - R = \frac{1}{1+\frac{V_0^2}{4E(E+V_0)}\sin^2(\frac{2a}{\hbar}\sqrt{2m(V_0+E)})}$$

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Finite Square Barrier

$V(x) = +V_0$ if $-a < x < a$, otherwise 0

$$k = \frac{\sqrt{2mE}}{\hbar} \qquad l = \frac{\sqrt{2m(V_0-E)}}{\hbar}$$

Tunneling

$$T = 1 - R = \frac{1}{1+\frac{V_0^2}{4E(V_0-E)}\sinh^2(\frac{2a}{\hbar}\sqrt{2m(V_0-E)})}$$

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Formalism

Bra-ket Notation

Ket: $\Psi(x,t) \leftrightarrow |\alpha\rangle = \alpha_1|e_1\rangle + \alpha_2|e_2\rangle + \alpha_3|e_3\rangle$

Conjugate Bra: $\Psi^*(x,t) \leftrightarrow \langle\alpha| = \alpha_1^*\langle e_1| + \alpha_2^*\langle e_2| + \alpha_3^*\langle e_3|$

Inner Products

$\langle T^\dagger\beta|\alpha\rangle = \langle\beta|T\alpha\rangle \qquad \langle\alpha|\beta\rangle = \langle\beta|\alpha\rangle^*$

Matrix of an Operator

$$[\hat{T}] = \begin{bmatrix} \langle e_1|\hat{T}|e_1\rangle & \langle e_1|\hat{T}|e_2\rangle & \langle e_1|\hat{T}|e_3\rangle \\ \langle e_2|\hat{T}|e_1\rangle & \langle e_2|\hat{T}|e_2\rangle & \langle e_2|\hat{T}|e_3\rangle \\ \langle e_3|\hat{T}|e_1\rangle & \langle e_3|\hat{T}|e_2\rangle & \langle e_3|\hat{T}|e_3\rangle \end{bmatrix}$$

Hermitian Operator

$T^\dagger = T \iff T_{mn} = T_{nm}^* \iff \langle\alpha|T\beta\rangle = \langle\alpha|T|\beta\rangle = \langle T^\dagger\alpha|\beta\rangle$

Uncertainty Principle

$\hat{Q}_1$, $\hat{Q}_2$ observables $\implies \sigma_{\hat{Q}1}\sigma_{\hat{Q}2} \geq |\frac{\langle[\hat{Q}_1,\hat{Q}_2]\rangle}{2i}|$

$$\sigma_{\hat{p}}\sigma_{\hat{x}} \geq \frac{\hbar}{2} \qquad \sigma_{\hat{E}}\sigma_{\hat{t}} \geq \frac{\hbar}{2}$$

Momentum-space Wavefunctions

$$\psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}px} \qquad \int_{-\infty}^{\infty} e^{i(k'-k)x}dx = 2\pi\delta(k'-k)$$

$$\Phi(p,t) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\psi(x,t)e^{-\frac{i}{\hbar}px}dx$$

Evolution Operator

$i\hbar\frac{d\vec{c}(t)}{dt} = [H]\vec{c}(t) \implies \vec{c}(t) = e^{-\frac{i}{\hbar}[\hat{H}]t}\vec{c}(0)$

Evolution Operator $\implies \hat{U}(t) = e^{-\frac{i}{\hbar}\hat{H}t}$ and $[U(t)] = e^{-\frac{i}{\hbar}[\hat{H}]t}$

$$\hat{Q}(t) = \hat{U}(t)^\dagger\hat{Q}\hat{U}(t) = e^{\frac{i}{\hbar}\hat{H}t}\hat{Q}e^{-\frac{i}{\hbar}\hat{H}t} \qquad i\hbar\frac{d\hat{Q}(t)}{dt} = [\hat{Q}(t),\hat{H}]$$

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Quantum Mechanics in 3D

$$i\hbar\frac{d\Psi}{dt} = -\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi \qquad \vec{p} = -i\hbar\nabla \qquad -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$$

$$[r_i, p_j] = -[p_j, r_i] = i\hbar\delta_{ij} \qquad [r_i, r_j] = [p_i, p_j] = 0$$

Angular Equation

$Y_0^0 = \left(\frac{1}{4\pi}\right)^{1/2}$	$Y_2^{\pm2} = \left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta e^{\pm2i\phi}$
$Y_1^0 = \left(\frac{3}{4\pi}\right)^{1/2}\cos\theta$	$Y_3^0 = \left(\frac{7}{16\pi}\right)^{1/2}(5\cos^3\theta - 3\cos\theta)$
$Y_1^{\pm1} = \mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta e^{\pm i\phi}$	$Y_3^{\pm1} = \mp\left(\frac{21}{64\pi}\right)^{1/2}\sin\theta(5\cos^2\theta - 1)e^{\pm i\phi}$
$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$	$Y_3^{\pm2} = \left(\frac{105}{32\pi}\right)^{1/2}\sin^2\theta\cos\theta e^{\pm2i\phi}$
$Y_2^{\pm1} = \mp\left(\frac{15}{8\pi}\right)^{1/2}\sin\theta\cos\theta e^{\pm i\phi}$	$Y_3^{\pm3} = \mp\left(\frac{35}{64\pi}\right)^{1/2}\sin^3\theta e^{\pm3i\phi}$

D.E. $\rightarrow sin(\theta)\frac{\partial}{\partial\theta}\left(\sin(\theta)\frac{\partial Y}{\partial\theta}\right) + \frac{\partial^2Y}{\partial\phi^2} = -l(l+1)\sin^2(\theta)Y$

$$Y_l^m(\theta,\phi) = \epsilon\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}}e^{im\phi}P_l^m(\cos\theta) = \Phi(\phi)\Theta(\theta)$$

$\epsilon = (-1)^m$ for m ≥ 0, $\epsilon = 1$ otherwise

$l = 0, 1, 2, ... \qquad m = -l, -(l-1), ..., l$

Ass. Legendre Funcs. $\implies P_l^m(x) = (1-x^2)^{|m|/2}\left(\frac{d}{dx}\right)^{|m|}P_l(x)$

Legendre Polynomials $\implies P_l(x) = \frac{1}{2^l l!}\left(\frac{d}{dx}\right)^l(x^2-1)^l$

Radial Equation

IMPORTANT NOTE: $u(r) = rR(r)$

D.E. $\rightarrow -\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \underbrace{\left[V + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right]}_{V_{eff}}u = Eu$

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$$R_{10} = 2a^{-3/2}\exp(-r/a)$$

$$R_{20} = \frac{1}{\sqrt{2}}a^{-3/2}\left(1-\frac{1}{2}\frac{r}{a}\right)\exp(-r/2a)$$

$$R_{21} = \frac{1}{\sqrt{24}}a^{-3/2}\frac{r}{a}\exp(-r/2a)$$

$$R_{30} = \frac{2}{\sqrt{27}}a^{-3/2}\left(1-\frac{2}{3}\frac{r}{a}+\frac{2}{27}\left(\frac{r}{a}\right)^2\right)\exp(-r/3a)$$

$$R_{31} = \frac{8}{27\sqrt{6}}a^{-3/2}\left(1-\frac{1}{6}\frac{r}{a}\right)\left(\frac{r}{a}\right)\exp(-r/3a)$$

$$R_{32} = \frac{4}{81\sqrt{30}}a^{-3/2}\left(\frac{r}{a}\right)^2\exp(-r/3a)$$

$$R_{40} = \frac{1}{4}a^{-3/2}\left(1-\frac{3}{4}\frac{r}{a}+\frac{1}{8}\left(\frac{r}{a}\right)^2-\frac{1}{192}\left(\frac{r}{a}\right)^3\right)\exp(-r/4a)$$

$$R_{41} = \frac{\sqrt{5}}{16\sqrt{3}}a^{-3/2}\left(1-\frac{1}{4}\frac{r}{a}+\frac{1}{80}\left(\frac{r}{a}\right)^2\right)\frac{r}{a}\exp(-r/4a)$$

$$R_{42} = \frac{1}{64\sqrt{5}}a^{-3/2}\left(1-\frac{1}{12}\frac{r}{a}\right)\left(\frac{r}{a}\right)^2\exp(-r/4a)$$

$$R_{43} = \frac{1}{768\sqrt{35}}a^{-3/2}\left(\frac{r}{a}\right)^3\exp(-r/4a)$$

Infinite Spherical Well

$$V(r) = \begin{cases} 0, & r < a \\ \infty, & r > a \end{cases} \qquad k = \frac{\sqrt{2mE}}{\hbar}$$

$$u(r) = Arj_l(kr) + Brn_l(kr) \xrightarrow{\text{origin}} R(r) = Aj_l(kr)$$

$$k = \frac{1}{a}\beta_{nl} \implies E_{nl} = \frac{\hbar^2}{2ma^2}\beta_{nl}^2$$

$$j_l(x) = (-x)^l\left(\frac{1}{x}\frac{d}{dx}\right)^l\frac{\sin(x)}{x} \qquad n_l(x) = -(-x)^l\left(\frac{1}{x}\frac{d}{dx}\right)^l\frac{\cos(x)}{x}$$

Hydrogen Atom

$$E_n = -\left[\frac{m}{2\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2\right]\frac{1}{n^2} = \frac{-13.6\text{eV}}{n^2} \qquad V(r) = \frac{-e^2}{4\pi\epsilon_0}\frac{1}{r}$$

$$\text{Alt. } E_n = -\frac{m_e c^2\alpha^2}{2n^2} \qquad V(r) = \frac{-hc\alpha}{r} \qquad \alpha = \frac{1}{4\pi\epsilon_0}\frac{e^2}{\hbar c} \approx \frac{1}{137}$$

$$R_{nl}(r) = \frac{1}{r}\rho^{l+1}e^{-\rho}v(\rho) \qquad \rho = \frac{r}{an} \qquad a = \frac{4\pi\epsilon_0\hbar^2}{me^2}$$

$$v(\rho) \sim L_{n-l-1}^{2l+1}(2\rho) \qquad \deg(v) = n-l-1$$

$$L_{q-p}^p(x) = (-1)^p\left(\frac{d}{dx}\right)^p L_q(x) \qquad L_q(x) = e^x\left(\frac{d}{dx}\right)^q(e^{-x}x^q)$$

$$\psi_{nlm} = \sqrt{\left(\frac{2}{na}\right)^3\frac{(n-l-1)!}{2n[(n+l)!]^3}}e^{-r/na}\left(\frac{2r}{na}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2r}{na}\right)Y_l^m(\theta,\phi)$$

Hydrogen Spectrum

$$E_\gamma = E_i - E_f = -13.6\text{eV}\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right) \qquad E_\gamma = h\nu = \frac{hc}{\lambda}$$

Angular Momentum

$$\vec{L} = \vec{r}\times\vec{p} \quad L_x = -i\hbar\left(y\frac{\partial}{\partial z}-z\frac{\partial}{\partial y}\right) \quad L_y = -i\hbar\left(z\frac{\partial}{\partial x}-x\frac{\partial}{\partial z}\right)$$

$$L_z = -i\hbar\left(x\frac{\partial}{\partial y}-y\frac{\partial}{\partial x}\right) \qquad \text{(Cyclic) } [L_x,L_y] = i\hbar L_z$$

$$L^2 = L_x^2+L_y^2+L_z^2 \qquad [L^2,L_x] = [L^2,L_y] = [L^2,L_z] = 0$$

$$[L_z,x] = i\hbar y \qquad [L_z,y] = -i\hbar x \qquad [L_z,z] = 0$$

$$[L_z,p_x] = i\hbar p_y \qquad [L_z,p_y] = -i\hbar p_x \qquad [L_z,p_z] = 0$$

Ladder Operators

$$L_\pm = L_x \pm iL_y \qquad [L_z,L_\pm] = \pm\hbar L_\pm \qquad [L^2,L_\pm] = 0$$

$$L^2 = L_\pm L_\mp + L_z^2 \mp \hbar L_z \quad L^2Y_m^l = \hbar^2 l(l+1)Y_l^m \quad L_zY_m^l = \hbar mY_l^m$$

$$L_\pm Y_l^m = \hbar\sqrt{l(l+1)-m(m\pm1)}Y_l^{m\pm1}$$

$$l = 0, 1/2, 1, 3/2, ... \qquad m = -l, -(l-1), ..., (l-1), l$$

Angular Momentum Operators in Spherical Coordinates

$$L_z = -i\hbar\frac{\partial}{\partial\phi} \qquad L_\pm = \pm e^{\pm i\phi}\left(\frac{\partial}{\partial\theta}\pm i\cot(\theta)\frac{\partial}{\partial\phi}\right)$$

$$L_x = -i\hbar\left(-\sin(\phi)\frac{\partial}{\partial\theta}-\cos(\phi)\cot(\theta)\frac{\partial}{\partial\phi}\right)$$

$$L_y = -i\hbar\left(\cos(\phi)\frac{\partial}{\partial\theta}-\sin(\phi)\cot(\theta)\frac{\partial}{\partial\phi}\right)$$

Spin

$$S^2|sm\rangle = \hbar^2 s(s+1)|sm\rangle \qquad S_z|sm\rangle = \hbar m|sm\rangle \qquad [S_x,S_y] = i\hbar S_z$$

$$S_\pm|sm\rangle = S_x \pm iS_y \qquad S_\pm = \hbar\sqrt{s(s+1)-m(m\pm1)}|s(m\pm1)\rangle$$

$$s = 0, 1/2, 1, 3/2, ... \qquad m = -s, -s+1, ..., s-1, s$$

Spin 1/2 Particles

$$\chi_+ = \begin{pmatrix}1\\0\end{pmatrix} \qquad \chi_- = \begin{pmatrix}0\\1\end{pmatrix} \qquad S^2\chi_\pm = \frac{3}{4}\hbar^2\chi_\pm \qquad S_z\chi_\pm = \pm\frac{\hbar}{2}\chi_\pm$$

$$S_\mp\chi_\pm = \hbar\chi_\pm \qquad S_\pm\chi_\pm = 0 \qquad S_x\chi_\pm = \frac{\hbar}{2}\chi_\mp \qquad S_y\chi_\mp = \mp\frac{\hbar}{2i}\chi_\mp$$

Spin 1/2 Matrix Operators and Pauli Matrices

$$[S^2] = \frac{3}{4}\hbar^2\begin{pmatrix}1&0\\0&1\end{pmatrix} \qquad [S_+] = \hbar\begin{pmatrix}0&1\\0&0\end{pmatrix} \qquad [S_-] = \hbar\begin{pmatrix}0&0\\1&0\end{pmatrix}$$

$$[S_x] = \frac{\hbar}{2}\underbrace{\begin{pmatrix}0&1\\1&0\end{pmatrix}}_{\sigma_x} \qquad [S_y] = \frac{\hbar}{2}\underbrace{\begin{pmatrix}0&-i\\i&0\end{pmatrix}}_{\sigma_y} \qquad [S_z] = \frac{\hbar}{2}\underbrace{\begin{pmatrix}1&0\\0&-1\end{pmatrix}}_{\sigma_z}$$

Spin Particles in Magnetic Fields

Schrodinger: $i\hbar\frac{\partial\chi}{\partial t} = \hat{H}\chi$

$\hat{H} = \hat{H}(\hat{x},\hat{p}) - \gamma_e\vec{S}\cdot\vec{B} \implies \hat{H} = \hat{H}(\hat{x},\hat{p}) - \gamma_e[S_z]B$

$\chi(t) = c_{1/2}(0)e^{\frac{-iE+\gamma_e\hbar B/2}{\hbar}t}\chi_+ + c_{-1/2}(0)e^{\frac{-iE-\gamma_e\hbar B/2}{\hbar}t}\chi_-$

Larmor Frequency $\implies \omega = \gamma_e B$

Matrix Form

$H = -\frac{\gamma_e B\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$ $\qquad \hat{H}\chi_\pm = E_\pm\chi_\pm = \mp(\gamma_e B\hbar)/2\cdot\chi_\pm$

$\chi(t) = \begin{pmatrix}\cos(\alpha/2)e^{i\gamma_e Bt/2}\\ \sin(\alpha/2)e^{-i\gamma_e Bt/2}\end{pmatrix}$ $\qquad \langle S_z\rangle = \frac{\hbar}{2}\cos(\alpha)$

$\langle S_x\rangle = \frac{\hbar}{2}\sin(\alpha)\cos(\gamma Bt)$ $\qquad \langle S_y\rangle = -\frac{\hbar}{2}\sin(\alpha)\sin(\gamma Bt)$

Stern-Gerlach Experiment

$F_z = \gamma_e\alpha S_z$ $\qquad B(x,y,z) = -\alpha x\hat{i} + (B_0+\alpha z)\hat{k}$

$H(t) = -\gamma(B_0+\alpha z)S_z$ for $0\le t\le T$

$\chi(t) = a\chi_+e^{-iE_+t/\hbar} + b\chi_-e^{-iE_-t/\hbar}$ $\qquad E_\pm = \mp\gamma(B_0+\alpha z)\frac{\hbar}{2}$

$\chi(t) = \left(ae^{i\gamma TB_0/2}\chi_+\right)e^{i(\alpha\gamma T/2)z} + \left(be^{-i\gamma TB_0/2}\chi_-\right)e^{-i(\alpha\gamma T/2)z}$

Other Useful Identities

Virial Theorem

$\frac{d}{dt}\langle\vec{r}\cdot\vec{p}\rangle = 2\langle T\rangle - \langle\vec{r}\cdot\nabla V\rangle$

Commutator Identities

$[A,BC] = [A,B]C + B[A,C]$ $\qquad [AB,C] = A[B,C] + [A,C]B$

$[f(x),p] = i\hbar\frac{df}{dx}$

Table of Integrals

Trigonometry:

$$\sin(a\pm b) = \sin a\cos b \pm \cos a\sin b$$
$$\cos(a\pm b) = \cos a\cos b \mp \sin a\sin b$$

Law of cosines:

$$c^2 = a^2 + b^2 - 2ab\cos\theta$$

Integrals:

$$\int x\sin(ax)\,dx = \frac{1}{a^2}\sin(ax) - \frac{x}{a}\cos(ax)$$
$$\int x\cos(ax)\,dx = \frac{1}{a^2}\cos(ax) + \frac{x}{a}\sin(ax)$$

Exponential integrals:

$$\int_0^\infty x^n e^{-x/a}\,dx = n!\,a^{n+1}$$

Gaussian integrals:

$$\int_0^\infty x^{2n}e^{-x^2/a^2}\,dx = \sqrt{\pi}\,\frac{(2n)!}{n!}\left(\frac{a}{2}\right)^{2n+1}$$
$$\int_0^\infty x^{2n+1}e^{-x^2/a^2}\,dx = \frac{n!}{2}a^{2n+2}$$

Hyrogen Orbitals

